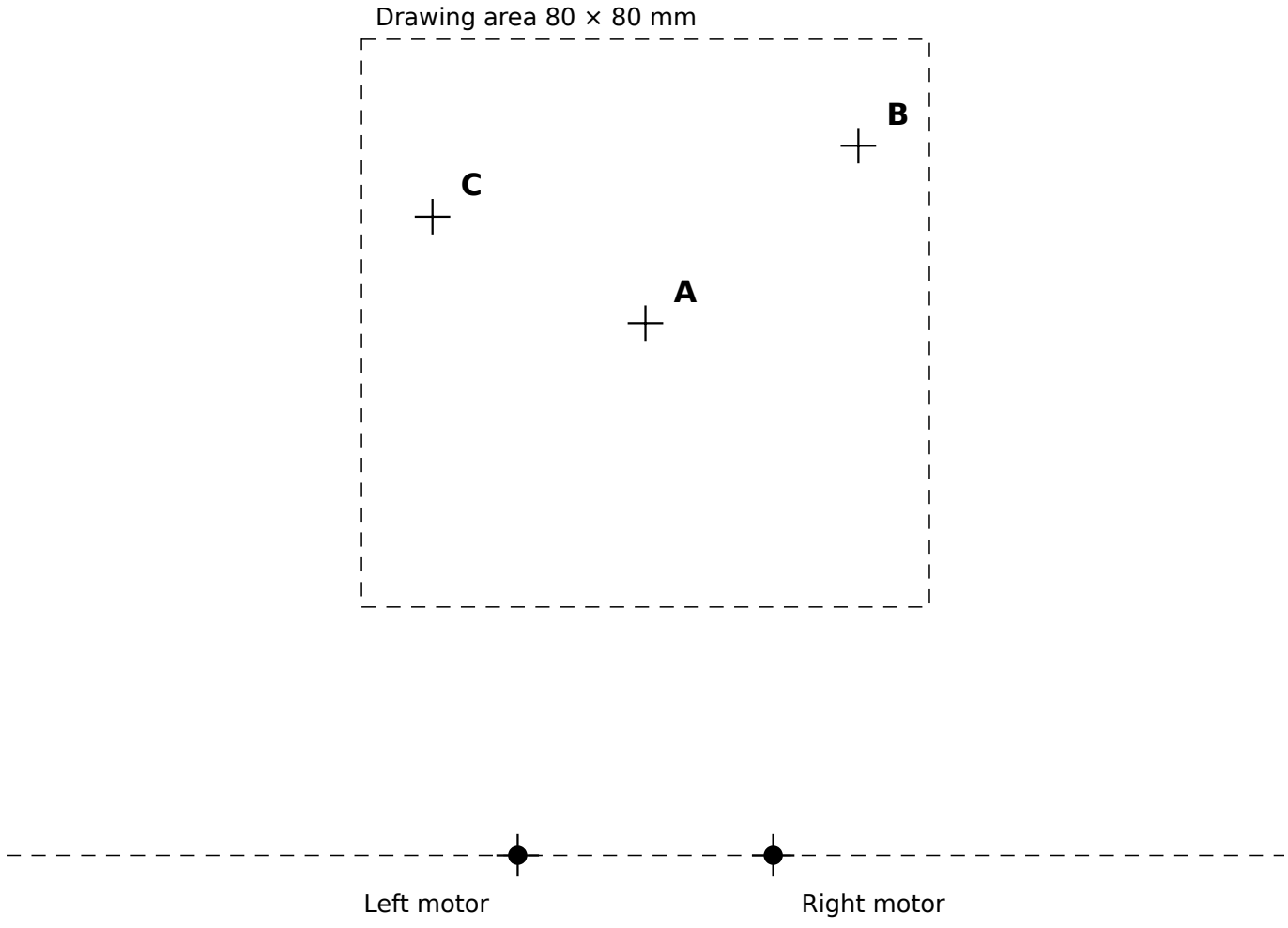
Worksheet: find the motor angles with a compass and protractor

PlotBot from above, scale 1:1. Upper arm 65 mm, forearm 90 mm, motors 36 mm apart.

1. Set the compass to 65 mm (upper arm). Draw a circle around each motor.
2. Set the compass to 90 mm (forearm). Put the point on target A and draw an arc that crosses both motor circles.
3. The arc crosses each motor circle twice. On each circle mark the outer crossing point (the left one for the left motor): that's the elbow.
4. Join them up with a ruler: motor – elbow – target.
5. Measure the angle at each motor with zero pointing right. Write it in the table. Then do the same for B and C.

Target	Left motor	Right motor
A		
B		
C		



Check: 50 mm